



*Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)*

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# **SMART CROP STRESS DETECTION SYSTEM**

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**The domain of the Project:  
EMBEDDED SYSTEMS AND IOT**

**GUIDE  
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**Period of the project  
Jan 2026 to March 2026**



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## Declaration

The project titled “SMART CROP STRESS DETECTION SYSTEM” has been mentored by Ms. Mehak Majeed, organised by SURE Trust, from Jan 2026 to March 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Done By:

Miss. SNEHA GORLA  
Signature

Mentor's Name:

Ms. Mehak Majeed  
Embedded System Engineer, ATFAAL Innovations

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## Abstract

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Agricultural productivity is highly dependent on timely irrigation and early detection of crop stress conditions. This project presents a **Smart Crop Stress Detection System** based on **Embedded Systems and IoT**, designed to predict crop stress and optimize water usage. The system uses an **ESP32 microcontroller** interfaced with sensors to monitor key parameters such as **soil moisture, temperature, humidity, and light intensity**. Based on predefined threshold values, the system classifies crop conditions into **Normal, Warning, and Critical** levels, enabling predictive monitoring rather than reactive response.

The proposed system features a **real-time IoT dashboard** that allows farmers to remotely monitor crop conditions and irrigation status. When critical stress conditions are detected, an **irrigation motor is automatically activated** through a relay module to reduce crop stress, with provisions for manual control. This solution helps conserve water, improve crop health, and provides a scalable and cost-effective approach for smart agriculture applications.



### Project Objectives

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The primary objective of this project is to design and implement a **Smart Crop Stress Detection System** using **Embedded Systems and Internet of Things (IoT)** technologies to support efficient water management in agriculture. The system aims to continuously monitor critical environmental and soil parameters affecting crop health and predict stress conditions at an early stage, thereby enabling timely corrective actions and minimizing water wastage.

Another important objective is to develop a **rule-based stress prediction mechanism** that classifies crop conditions into **Normal, Warning, and Critical** levels based on predefined threshold values. This predictive approach helps farmers understand upcoming stress conditions rather than reacting after crop damage has occurred. The system is intended to analyze real-time data from sensors such as soil moisture, temperature, humidity, and light intensity to determine the irrigation requirements of crops accurately.

The project also aims to implement an **automated irrigation control system** using actuators such as motors and relay modules. When critical stress conditions are detected, the system automatically activates irrigation to restore optimal soil moisture levels while preventing over-irrigation. Additionally, a manual override feature is included to allow users to control the irrigation system when necessary.

Another key objective is to design a **user-friendly IoT dashboard** that provides real-time visualization of sensor data, crop stress status, and motor operation. This dashboard enables remote monitoring and improves decision-making for farm owners. The project further aims to ensure **scalability, reliability, and cost-effectiveness**, making the system suitable for small to medium-scale agricultural fields. Future enhancements such as weather forecasting integration and advanced analytics can be incorporated to improve accuracy. .



## Components

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### Hardware Components

1. **ESP32 Development Board**  
Acts as the central controller to process sensor data, execute stress prediction logic, control actuators, and manage communication.
2. **Soil Moisture Sensor**  
Measures soil water content to detect crop water stress and decide irrigation requirements.
3. **Temperature and Humidity Sensor (DHT11 / DHT22)**  
Monitors environmental conditions influencing crop stress and evaporation rate.
4. **Light Intensity Sensor (LDR)**  
Measures sunlight exposure affecting crop growth and heat stress.
5. **Relay Module**  
Provides electrical isolation and allows the ESP32 to safely control the irrigation motor.
6. **DC Motor / AC Motor (Irrigation Motor)**  
Used to control the irrigation process when critical crop stress conditions are detected.
7. **Power Supply (5V / 12V Adapter or Battery)**  
Supplies stable power to the ESP32, sensors, GSM module, and motor.

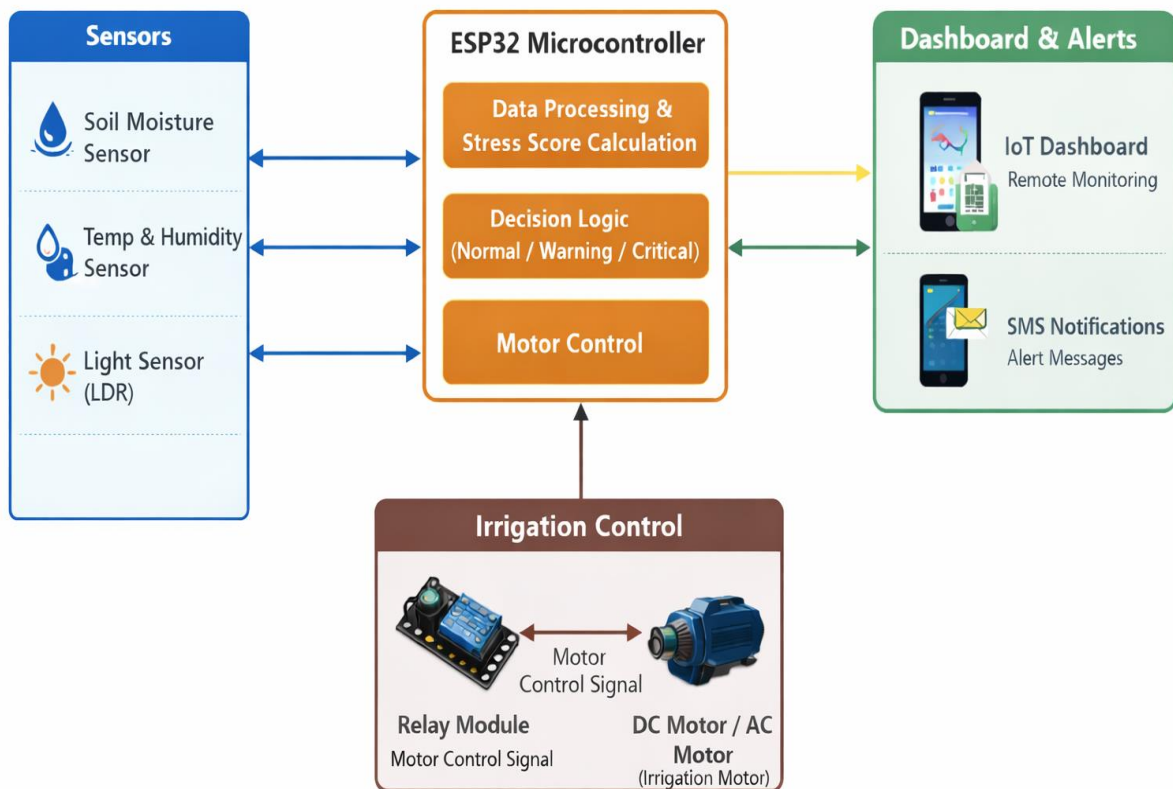
### Software Components:

- **VS Code with PlatformIO / Arduino IDE**
  - Used to write, compile, and upload the ESP32 program.
- **ESP32 Board Support Package and Libraries**
  - Required for sensor interfacing, Wi-Fi communication, and motor/relay control.
- **Sensor Libraries (DHT, ADC, etc.)**
  - Provide functions to read and process data from soil moisture, temperature, **humidity**, and light sensors.
- **IoT Dashboard Platform (Blynk / Web Dashboard)**
  - Displays real-time sensor data, crop stress status (Normal / Warning / Critical), and motor ON/OFF status.
- **Wi-Fi Communication Libraries**
  - Enable ESP32 to send sensor data to the dashboard in real-time.



## Block Diagram

### Smart Crop Stress Detection System





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## **Methodology**

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The Smart Crop Stress Detection System is designed using an **Embedded Systems and IoT-based architecture** to continuously monitor crop and environmental conditions, predict crop stress, and automatically manage irrigation. The methodology begins with the **collection of real-time data** from multiple sensors deployed in the agricultural field. Sensors such as the soil moisture sensor, temperature and humidity sensor, and light intensity sensor are interfaced with an **ESP32 microcontroller**, which acts as the central processing unit of the system. These sensors periodically measure parameters that directly influence crop health and water requirements.

The collected sensor data is processed within the controller to calculate a **Crop Stress Score (CSS)**. Each parameter is normalized and assigned a predefined weight based on its importance to crop stress, with soil moisture given the highest priority. The weighted values are combined to compute the overall stress score, which reflects the current condition of the crop. Based on the calculated score, the crop condition is classified into **Normal, Warning, or Critical** stress levels. This quantitative evaluation enables the system to predict stress conditions rather than responding only after severe crop damage occurs.

Once the stress level is determined, the ESP32 executes the **decision-making logic**. In the Normal state, the system continues monitoring without intervention. In the Warning state, the system sends an **SMS alert** to the farmer indicating potential stress conditions. When the stress level reaches the Critical state, the controller automatically activates the **irrigation motor** through a relay module to restore optimal soil moisture levels. Simultaneously, SMS alerts are sent to notify the farmer of the critical condition and irrigation activity. The motor is turned off once safe moisture levels are reached to prevent over-irrigation and water wastage.

In addition to automation, the system integrates an **IoT-based dashboard** that provides real-time visualization of sensor readings, crop stress status, and motor operation. Communication between the ESP32 and the dashboard is achieved using Wi-Fi connectivity, allowing farmers to monitor their fields remotely through a mobile application. Manual control options are also provided via the dashboard for enhanced flexibility. This methodology ensures efficient water usage, early stress prediction, and improved crop management while maintaining scalability and reliability for future enhancements.